# Module 01: Lab 01

DSE6211OM_SU2024S4_Machine Learning+Artificial Intelligence

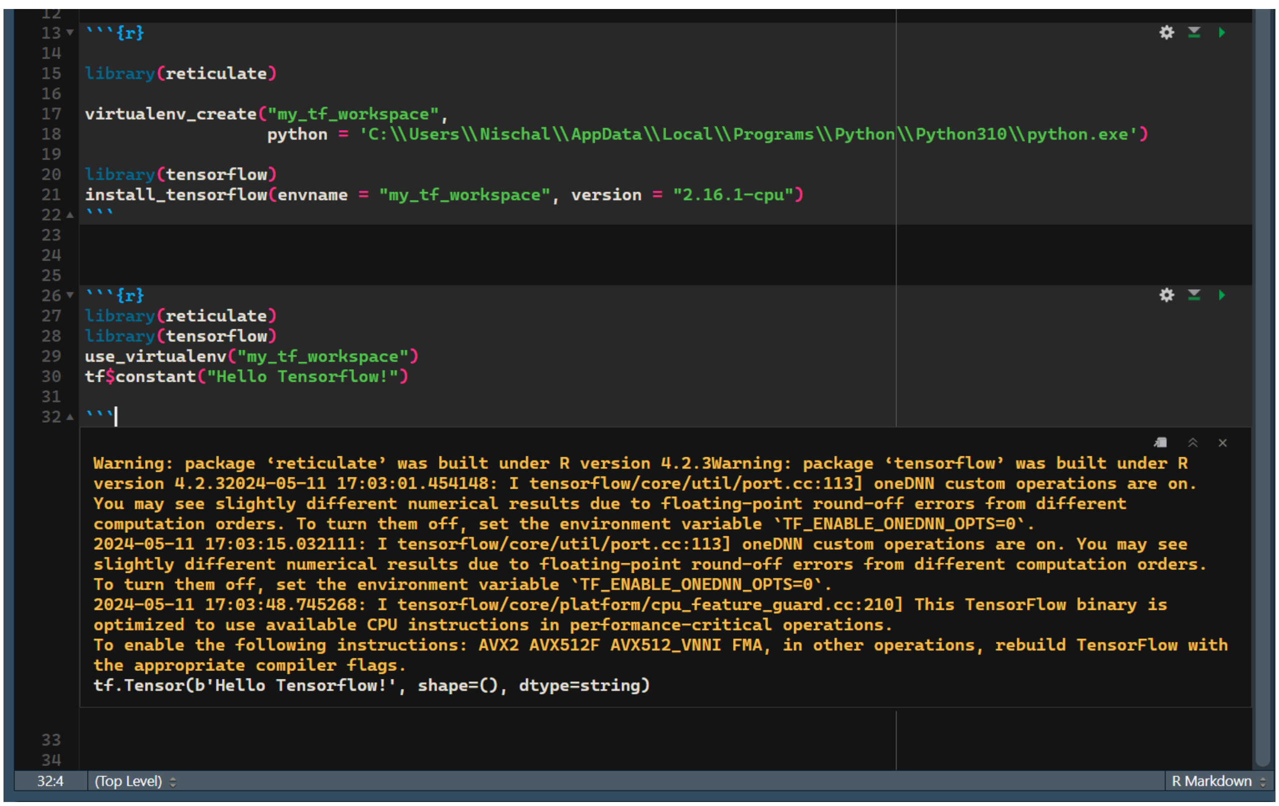
Nischal Pant

**1) What is the main difference between supervised and unsupervised learning? Was the type of learning performed in the 'First Keras Example' section supervised or unsupervised learning? Why?**

The main difference between them is that supervised learning uses labeled data and requires human input. This allows the model to make predictions and using labeled inputs and outputs, the model can measure its accuracy and learn over time.

In the first Keras example, we used labeled input and output data, we are trying to predict mpg, using the cylinders of the engine, displacement, and hp of the car. This makes the model supervised.

**2) Take a screenshot of the output of 'tf$constant("Hello Tensorflow!")' as described above.**



~~~~
```{r}

library(reticulate)

virtualenv_create("my_tf_workspace",
                  python = 'C:\\Users\\Nischal\\AppData\\Local\\Programs\\Python\\Python310\\python.exe')

library(tensorflow)
install_tensorflow(envname = "my_tf_workspace", version = "2.16.1-cpu")
```



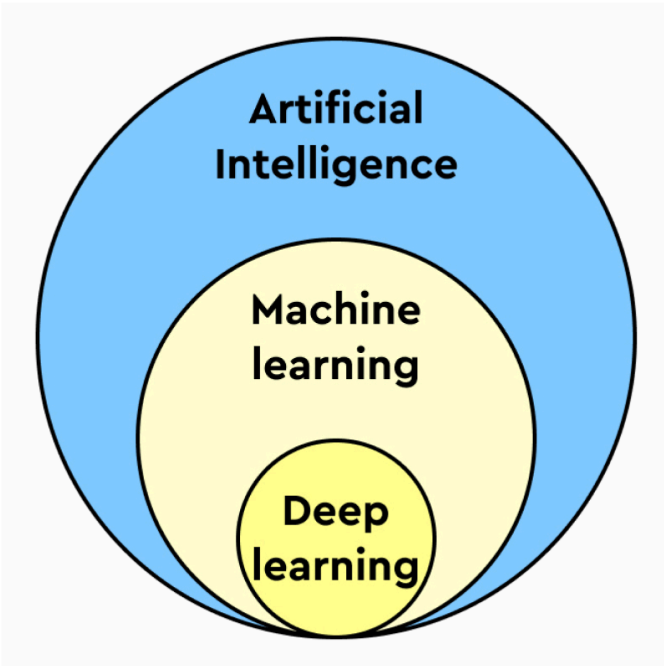
```{r}
library(reticulate)
library(tensorflow)
use_virtualenv("my_tf_workspace")
tf$constant("Hello Tensorflow!")

```

Warning: package 'reticulate' was built under R version 4.2.3Warning: package 'tensorflow' was built under R
version 4.2.32024-05-11 17:03:01.454148: I tensorflow/core/util/port.cc:113] oneDNN custom operations are on.
You may see slightly different numerical results due to floating-point round-off errors from different
computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2024-05-11 17:03:15.032111: I tensorflow/core/util/port.cc:113] oneDNN custom operations are on. You may see
slightly different numerical results due to floating-point round-off errors from different computation orders.
To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2024-05-11 17:03:48.745268: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is
optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: AVX2 AVX512F AVX512_VNNI FMA, in other operations, rebuild TensorFlow with
the appropriate compiler flags.
tf.Tensor(b'Hello Tensorflow!', shape=(), dtype=string)
~~~~



I had to use a different version of TensorFlow as 2.8 was not working and 2.16.1 was the latest one that was compatible. I also had Python 3.11 which i had to downgrade to 3.10.

**3) Briefly describe the fields of machine learning and deep learning, as well as the main difference(s) between both fields.**




Machine learning and deep learning all fall under the generalized term of artificial intelligence. Artificial learning is a branch of computer science where computational models perform tasks that humans only could do, such as reasoning, making decisions, and solving problems.

Machine learning involves using data (training and testing) to make predictions and or decisions without human input (there are exceptions).

Deep learning is a subset of machine learning and involves neural networks that can automatically learn hierarchical representations of data.

Machine learning itself has many more algorithms and processes other than neural networks.

**4) What type of transformation is performed by each layer in a neural network? What is the purpose of the loss function in training a neural network?**

There are 3 layers in a neural network:

**Input Layer:** This is simply data inputting and is considered the entry point into the neural network. There is no transformation performed in the layer.

**Hidden layer:** This is the layer of neural networks that can be composite and can have many layers. The transformations are nonlinear and each layer has a weighted sum that transforms the input value and gives out outputs.

**Output layer:** This is the final output layer and the transformation is dependent on the layers that are within the network. The more layers, the more complex.

**The loss function:** the function's purpose is to measure the accuracy of the network which it does so by comparing the true labels to the training data. This can measure the model's performance and can help optimize the model. The model wants to keep this value low as possible to increase the precision.